Brianne Giglio

Lab 6 Day 3: Conversion of Borneol to Camphor

November 28, 2023

CHEM 2081-004

Exp. No. 6D3	Experiment/Subject	Date	41
Name Brianne Giglio	Lab Partner	Locker/ Desk No.	Course & Section No.

1) In 10 mL round bottom flask add 0.3 g borneal product w/ 0.3 mL glacial acetic acid. disolve in 1 mL acetone. add boiling chip

✓ added 0.3 g
-had to use some extra mine dissapeared

2) assemble reflux with distillation column and place in water bath 40-50°C If borneol does not dissolve, add 0.5 mL acetone to mixture

✓

3) once all dissolved, add 6 mL aq bleach. should be done through top of condensor dropwise for 35 mins. swirl after each addition, allow to reflux for 15 mins after all is added

✓ as bleach added, two layers begin to form.

4) let mixture cool to room temp and extract camphor w/ 2 2 mL portions of methylene chloride. use centrifuge tube

✓ extracted top layer

5) neutralize w/ ~2mL saturated sodium bicarbonate. the prescense of acidic residuals will be determined by production of CO_2 bubbles after every addition

✓

6) wash w/ 2 mL 5% sodium bisulfate and 2 mL water

7) dry w/ anhydrous sodium sulfate

8) evaporate DCM over water bath 50°C. weigh collected crystals.

✓ watch glass weighs: 33.28 g

weight w/ product 33.49 g

weight product: 0.21 g

Signature	Date	Witness/TA	Date

STUDENT LAB NOTEBOOK Note: Place fold-over back cover under copy sheet before writing

Introduction

In this lab, borneol was converted into camphor using glacial acetic acid and aqueous sodium hypochloride.

Materials and Methods

In a long-neck round-bottom flask, 0.3 g of borneol product from week 2 was combined with 0.3 mL of glacial acetic acid. The mixture was dissolved in 1 mL of acetone. Next, the distillation column was assembled for reflux. It was placed in a water bath with a temperature between 40-50℃. Once all of the reagents had dissolved, 6 mL of aqueous sodium hypochloride solution was added to the mixture slowly through the top of the condenser. The reaction was then left to continue for 15 minutes. Next, the mixture was allowed to cool to room temperature. The camphor that was produced was extracted using two 2 mL portions of DCM using a centrifuge tube. The top layer was removed each time. The solution was then neutralized using 2 mL of saturated sodium bicarbonate. Next, the organic solution was washed with 2 mL of 5% sodium bisulfate and 2 mL of water. The solution was then dried with anhydrous sodium sulfate. The DCM was evaporated over a water bath and the produced crystals were weighed.

Results

Weight of Watch Glass	33.28 g
Weight of Watch Glass and Product	33.49 g
Weight of Product	0.21 g

Table 1: Weight of watch glass, weight of watch glass and product, weight of product

Calculations

Weigh of Product: $33.49 - 33.28 = 0.21\ g\ camphor$

Theoretical Yield:

$$0.30\ g\ borneol\ \times \frac{1\ mol}{154.25\ g} \times \frac{1\ mol}{1\ mol} \times \frac{152.23\ g}{1\ mol} = 0.30\ g\ camphor$$

Percent Yield:

$$\frac{0.21\ g}{0.30\ g} \times 100 = 70\%$$

Discussion

This experiment was successful in creating camphor. The bottom layer in all of the washes was the organic layer because DCM has a density of 1.33 g/mL which is more than that of water which proves the product would have been on the bottom. This can be confirmed because solid crystals were left behind when the DCM was evaporated from the product. If no product had been formed, the solution would have evaporated when placed on heat but because a solid was produced, it can be confirmed that camphor was formed. There are some sources of error that could have led to there only being a 70% yield. The first is that the product from week two was not completely pure there could have been some remnants of water left which could have caused a lower percent yield. Another possibility is that the reaction was not given enough time to completely finish. This would cause an incomplete reaction and therefore a lower yield. To solve this issue, the reaction could be left for longer.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.

Lab Performance

Exp 6 D3

Brianne Giglio — H

Name: *Section:* *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ○ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* [ ]